PathGuard: Discrete Logic-Driven Obstacle Avoider

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I. Abstract

This project presents the design and implementation of 'PathGuard: Discrete Logic-Driven Obstacle Avoider, an autonomous robot car developed using fundamental hardware logic principles without the use of a microcontroller. The system detects obstacles in its path using infrared (IR) sensors and navigates by steering in the opposite direction, driven entirely by discrete transistor-based NOT gates and a dual H-bridge motor driver. The motivation behind this work stems from the long-term goal of developing assistive technologies for the elderly and physically challenged. The physical requirement for such assistive robots is the ability to navigate indoor environments autonomously, avoiding common household obstacles. The objective of this project is to create a simplified, cost-effective hardware prototype that demonstrates logic-based autonomous navigation using only analog electronics and combination logic. The robot achieves this by interpreting sensor signals through hardware-level inversion and using them to control the direction of two independently driven DC motors. This prototype serves as a fundamental module for the future development of human follow-up robots, with potential applications in elderly care, indoor mobility assistance, and autonomous service robotics. By demonstrating practical deployment of RTL logic and digital motor control principles, the project bridges theoretical learning with real-world application and provides a scalable foundation for further research and system integration.

Keywords: Obstacle Avoidance, Discrete Logic, RTL NOT Gate, L298N Motor Driver, Infrared Sensors, Hardware Automation

II. Implementation

This section provides a comprehensive explanation of the design and construction of the obstacle-avoiding robot. The project was executed without the use of a microcontroller, relying instead on discrete digital logic, transistor-based inversion, and a dual H-bridge motor driver. The implementation process began with a review of relevant electrical and logical foundations, followed by practical circuit assembly, simulation testing, and problem-solving during real-world deployment.

A. Theoretical Basis and Component Interaction

Obstacle avoidance in mobile robots relies on real-time environmental feedback. Most commonly, this is implemented using microcontrollers that process sensor data to dynamically adjust pathing algorithms [1], [2]. However, this project demonstrates that a similar functionality can be achieved through purely hardware-level logic. The core components used include infrared (IR) sensors, RTL NOT gates, and the L298N motor driver. IR sensors emit infrared light and detect reflected beams from nearby objects. A HIGH (1) signal indicates no obstacle detected, whereas a LOW (0) signal indicates the presence of an object. These binary signals are ideal for digital decision-making circuits [3]. To complement the raw IR signal, a Resistor-Transistor Logic (RTL) NOT gate was used to provide an inverted version of the output. The purpose of this inversion is to always have both the original and its opposite logic available, simplifying motor direction control.The RTL inverter is built using a single NPN transistor and a resistor connected to its base. The IR sensor's output is passed through a base resistor to the base of the transistor. The emitter of the transistor is connected to ground (GND). The collector is connected to a pull-up resistor and then to +5V, and the inverted output is taken from the collector. When the IR sensor outputs HIGH (1) the transistor conducts and pulls the collector LOW (0) giving an inverted signal. When the IR sensor outputs LOW (0) the transistor remains off and the pull-up resistor keeps the collector HIGH (1). The NOT gate was constructed using an NPN transistor with a pull-up resistor, as illustrated in Table I.

TABLE I: RTL NOT Gate Logic Table

IR Output	Transistor State	Inverted Output
1 (HIGH)	ON	0 (LOW)
0 (LOW)	OFF	1 (HIGH)

This design ensures that for every sensor reading, both original and complementary signals are available. These signals are passed to the L298N motor driver, which functions as a dual H-Bridge and allows bidirectional control of each DC motor. The internal switching mechanism of the L298N is shown in Table II, explaining how motor direction is controlled based on INx inputs [4].

TABLE II: Motor Direction Table

INx	INy	OUTx	OUTy	Motor Direction
1	0	HIGH	LOW	Current flows: OUTx → OUTy
0	1	LOW	HIGH	Current flows: OUTy → OUTx
0	0	LOW	LOW	Motor off (no voltage)
1	1	HIGH	HIGH	Motor off (no potential difference)

In the robot design, OUT1/OUT2 control the left motor and OUT3/OUT4 control the right motor. It also has input pins to control the motor outputs IN1 and IN2 control OUT1 and OUT2 (left motor) and IN3 and IN4 control OUT3 and OUT4 (right motor) [2], [3]. The motor driver receives power from an external battery (Vcc) and also includes 5V and GND connections for logic and grounding. In terms of connection logic, the left IR sensor (both direct and inverted outputs) is connected to IN3 and IN4, controlling the right motor. Similarly, the right IR sensor (both direct and inverted outputs) is connected to IN1 and IN2, controlling the left motor. This crossed connection design ensures that when an obstacle is detected on one side, the opposite motor responds, allowing the robot to turn away from the obstacle or adjust its path accordingly. [1]

The robot's navigation behavior was defined using a truth table, presented in Table III. This table summarizes the input conditions and the resulting motor behavior.

TABLE III: Robot Navigation Truth Table

L IR	R IR	IN1	IN2	IN3	IN4	Action
1	1	1	0	1	0	Move Forward
0	1	1	0	0	1	Turn Right
1	0	0	1	1	0	Turn Left
0	0	0	1	0	1	Move Backward

B. Circuit Implementation and Setup

The physical circuit was assembled on a breadboard and tested in stages to ensure functional isolation and correct logic propagation. The RTL NOT gates were constructed using NPN transistors and tested individually using IR sensor outputs before full integration. The final system was mounted on a two-wheeled robot chassis powered by two 3.7V lithium-ion cells in series, delivering sufficient current for both the logic circuit and the motors. Motor direction was verified experimentally by testing the input combinations on the L298N motor driver to confirm expected rotational behavior. IR sensors were positioned at the front-left and front-right of the chassis, angled slightly outward for wider detection coverage. All signal and power wiring was routed on a breadboard fixed on the chassis. Fig. 1 shows the fully assembled robot with all key components: two IR sensors, the breadboard with RTL NOT gates, the L298N motor driver module, power supply, and motor connections. The sensor positioning and the central logic board layout are clearly visible.

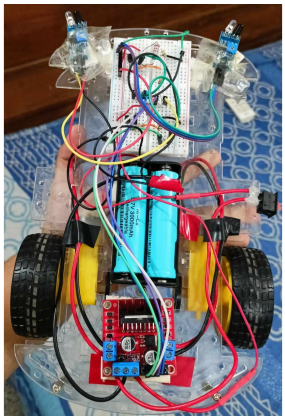


Fig. 1: Final robot build with IR sensors, RTL gates, and L298N on chassis

To document the internal circuit logic, a hand-drawn schematic was first produced during initial prototyping (Fig. 2). It maps the IR sensors' output to the RTL NOT gate, and then into the appropriate L298N driver inputs, showing both the control and power pathways.

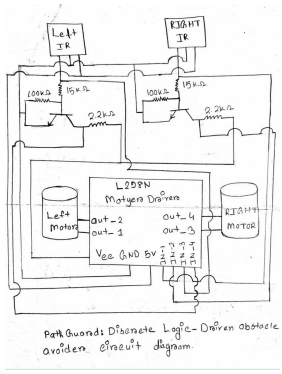


Fig. 2: Hand-drawn circuit schematic used in early prototyping

For clearer interpretation, a digital block-level wiring diagram was created in vector format (Fig. 3). This schematic illustrates the transistor inverters, sensor-motor pairing, and full electrical layout including the dual motor outputs and power bus routing.

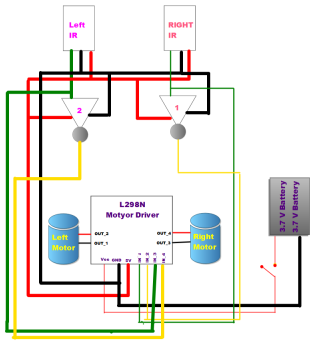


Fig. 3: Block Diagram used in Creating and Debugging

A breadboard simulation view was also designed (Fig. 4) using standard prototyping components with TinkerCAD. This model accurately reflects how the RTL NOT gates, IR sensors, and L298N chip are laid out on a physical breadboard. The design served as a reference during circuit debugging and testing.

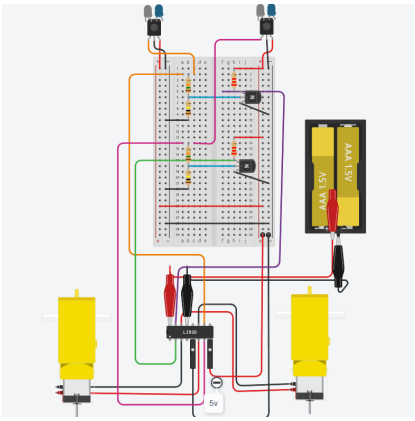


Fig. 4: CAD Diagram used for proper implementation and Debugging

Each of these illustrations supports the physical realization of the project's logic system, providing a visual breakdown of how IR sensor input is transformed through discrete RTL logic and drives real-time motor control using the L298N module.

C. Challenges and Solutions

During implementation, several technical challenges were encountered and systematically addressed. One of the primary issues was the limited sensitivity of the IR sensors. Thin obstacles, such as chair legs, occasionally passed undetected due to the narrow angle of the IR beam. This constraint is common in IR-based systems and may be mitigated in future versions by incorporating ultrasonic sensors, which provide a wider and more consistent detection range [2], [4]. Another notable challenge involved voltage fluctuations from the battery supply. As the battery discharged over time, the IR sensor calibration — which is set using a potentiometer — would shift, leading to inconsistent obstacle detection. This issue necessitated repeated manual adjustments during testing. To enhance reliability, the use of a voltage regulator or boost converter is recommended in future iterations to maintain a stable operating voltage. Lastly, the system initially faced motor direction mismatches caused by reversed wiring polarity between the L298N outputs and the DC motors. This was identified through trial-and-error testing and resolved by analyzing the direction of current flow and correcting the input logic accordingly. Despite these challenges, the robot ultimately demonstrated consistent and reliable performance in indoor environments, successfully avoiding obstacles based on real-time sensor feedback and logic-driven motor control.

III. Conclusion

To conclude, this project successfully achieved its core objective of designing and implementing a functional obstacle-avoiding robot entirely through hardware-based combinational logic, without the use of any microcontroller. The robot was constructed using infrared (IR) sensors, a dual H-Bridge L298N motor driver, and Resistor-Transistor Logic (RTL) NOT gates implemented with discrete components. Its ability to detect obstacles in real time and steer away using inverted control logic was verified through repeated indoor testing. The cross-motor logic and truth-table-based control system demonstrated accurate behavior in structured environments, validating the effectiveness of the underlying design methodology [1], [2]. In addition to confirming obstacle avoidance, the system also served as a hardware realization of digital logic design principles taught in CSE350, particularly in areas such as signal inversion, truth table construction, and discrete logic implementation using transistors. This aligns with previous research that emphasizes hardware-level decision-making in low-cost embedded systems [3]. However, the current model presents certain limitations. Thin objects such as chair legs were sometimes undetected due to the narrow angular detection range of the IR sensors — a limitation similarly observed in prior studies using reflective sensor-based systems [1], [4]. Furthermore, the system lacks variable speed control, and frequent battery voltage drops affected IR sensor calibration, requiring manual tuning. Future iterations of this system could benefit from multiple enhancements. Functional upgrades may include integrating ultrasonic sensors for wider distance detection, vision-based modules for target recognition and tracking, and IR sensor arrays for multi-directional feedback [2], [5]. Additionally, PWM speed control using hardware-only techniques could enable variable motor response without microcontroller intervention [3]. To improve reliability, the addition of voltage regulation components, such as low-dropout (LDO) regulators or boost converters, would ensure stable logic operation even under inconsistent power supply conditions. These improvements could expand the current robot into a foundation for more advanced systems, including human-following assistive robots in indoor environments — a research direction supported by recent developments in autonomous mobility and assistive robotics [1], [5].

References

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